

AI Literacy Glossary

Week 2 - How Machine Learning Works

Purpose. This glossary adds the vocabulary and concepts introduced in Week 2. It assumes the Week 1 definitions of artificial intelligence, machine learning, AI model, deep learning, representation, transfer and the wider AI-enabled system, so those terms are not redefined here.

Week 2 focus: how learning occurs, the main ways machines learn, how data and objectives shape the resulting model, how we test whether learning generalises, and why apparently strong patterns can still mislead.

Learning process and core mechanics

Database

A database is an organised collection of recorded information that can be stored, searched and retrieved. A database does not have to be intelligent: its main purpose is to keep information in a structured form so that particular records or values can be found later.

This matters in machine learning because a database and a learned model solve different problems. A database answers questions such as "What information was stored for this case?" A model applies learned patterns to a new case.

Example

A university database can store a student's enrolment, course results and contact details. Looking up the recorded final grade is retrieval, not machine learning.

Retrieval

Retrieval means finding and returning information that already exists in a database, document collection, website or other information source. The information is looked up rather than predicted from learned patterns.

Modern AI applications often combine retrieval with machine learning. For example, an application may retrieve a policy document and then ask a model to summarise it. The retrieval step and the model-generated step are conceptually different.

Retrieval asks, "What information is already there?" Machine learning asks, "Given what was learned, what should be predicted for this new case?"

Training

Training is the process in which data and a learning objective are used to shape a model. During training, the system repeatedly adjusts internal values called parameters so that its outputs improve according to some measure of success. The difference between a prediction and the desired or observed outcome can provide an error signal that helps guide those adjustments.

Training is therefore the stage in which the model acquires its task-specific behaviour. Using a trained model later is a different process called inference.

Example

A fraud-detection model may be trained using thousands of past transactions for which the eventual outcome is known.

Inference

Inference is the process of applying a trained model to a new input in order to produce a prediction or other output. Inference uses the learned model; it does not normally rebuild the model from scratch each time it is used.

The distinction is important because interacting with an AI system does not necessarily mean that the underlying model is learning from every interaction.

Example

After training, the fraud model receives a transaction made today and estimates whether it appears fraudulent. That use of the trained model is inference.

Objective function

An objective function is a measurable definition of what counts as better performance during learning. It gives the training process a direction: changes that improve the objective are preferred over changes that make it worse.

Objectives can be simple even when the real-world goal is complicated. Examples include reducing classification errors, reducing forecasting error, improving a ranking, or increasing a reward.

The objective is the compass that guides learning - but the compass points towards what was measured, not automatically towards everything humans ultimately care about.

Proxy objective

A proxy objective is a measurable target used in place of a broader goal that is difficult to measure directly. A proxy can be useful, but it can also distort behaviour when it captures only part of what people actually want.

A model can therefore optimise its proxy very successfully while still producing an undesirable real-world outcome.

Example

A recommendation system might optimise clicks or time spent because these are easy to measure, even though the broader goal is to provide useful or beneficial recommendations.

Optimisation

Optimisation is the process of changing a model so that it performs better according to its objective. Conceptually, the learning process repeatedly searches for adjustments that improve performance.

This course treats optimisation at a high level. The important sequence is: predict -> compare -> adjust -> repeat.

Convergence

Convergence is the stage of a learning process at which further adjustments produce little additional improvement according to the chosen objective.

Convergence does not mean that the model has discovered the truth, become unbiased, or will work reliably everywhere. A model can converge while learning from poor data or pursuing an inappropriate objective.

Major ways machines can learn

Supervised learning

Supervised learning uses examples for which a desired outcome is already provided. The system learns relationships between the input information and those known outcomes.

The word "supervised" does not mean that a human watches every training step. It means that the examples provide a target or answer against which learning can be guided.

Example

Past emails labelled "spam" or "not spam" can be used to train a classifier that predicts the label for a new email.

Label

A label is the outcome or category attached to an example in supervised learning. It tells the learning process what answer should be associated with that training case.

Labels can come from measurements, expert decisions, human annotation, surveys, institutional categories or historical outcomes. Labels used as the reference answer are sometimes called ground truth, but that phrase can overstate certainty: a label can contain uncertainty, disagreement or past human bias.

Feature / input variable

A feature, also called an input variable, is a piece of information made available to a model when it makes a prediction. A case may contain one feature or many features.

Features can be numerical, categorical, textual, visual or derived from other data. Which information is included influences which patterns a model can discover.

Example

For predicting house prices, features might include location, floor area, number of bedrooms and age of the building.

Target variable / outcome

The target variable is what a supervised-learning system is trying to predict. It is the outcome that the training examples provide and that the model later estimates for new cases.

Defining the target is a substantive human decision. A poorly chosen target can direct learning towards a result that is measurable but not truly useful.

Classification

Classification is a supervised-learning task in which the model predicts which category a case belongs to. The output is one of a defined set of classes.

Classification answers questions such as "Which kind is this?" or "Does this case belong to this category?"

Example

Spam versus not spam, diseased versus healthy, or one wildlife species versus another are classification problems.

Regression

Regression is a supervised-learning task in which the model predicts a numerical quantity rather than a category.

Regression answers questions such as "How much?", "How long?" or "What value?"

Example

Predicting energy demand, house price, waiting time or crop yield are regression problems.

Decision tree

A decision tree is an interpretable model that makes a prediction by following a sequence of learned decisions. Each branch asks a question about the input and routes the case towards another branch or a final prediction.

Decision trees are useful for teaching because they can resemble IF-THEN rules. The important difference from symbolic AI is that the branching rules are learned from data rather than written manually.

Unsupervised learning

Unsupervised learning looks for structure in data without being given known target answers for every example. The system is asked to discover patterns such as groups, similarities or unusual cases.

The central question changes from "Can you predict the supplied answer?" to "What structure appears to exist in these observations?"

Clustering

Clustering is an unsupervised-learning task that groups cases that are similar to one another according to the information available.

The resulting groups are not automatically meaningful categories. Humans still need to interpret what the clusters represent and whether they are useful.

Example

A retailer might cluster customers according to purchasing patterns and then examine whether the resulting groups correspond to useful customer segments.

Anomaly detection

Anomaly detection aims to identify cases that look unusual compared with the broader pattern in the data. An anomaly is not necessarily an error or a problem; it is simply something that differs from what is typical.

Anomaly detection is used in areas such as fraud screening, cybersecurity, equipment monitoring and scientific data analysis.

Self-supervised learning

Self-supervised learning creates a learning task from the data itself rather than requiring humans to manually label every example. One common idea is to hide or withhold part of the input and train the system to predict the missing part.

This makes it possible to learn from enormous collections of text, images or other data that would be impractical to label by hand. Week 3 develops how this idea connects to modern generative AI.

Reinforcement learning

Reinforcement learning is an approach in which a system learns through the consequences of actions. It receives feedback, often represented as a reward - a numerical signal indicating how desirable an outcome is - and gradually favours actions that tend to lead to better outcomes over time.

It is especially relevant when learning involves sequences of decisions rather than a single prediction.

Example

A game-playing system can try different actions, observe the eventual result, and learn which choices tend to increase its chance of winning.

Data and what the model gets to experience

Dataset

A dataset is an organised collection of examples used for analysis or machine learning. A dataset may be stored in a spreadsheet, database, image collection, document archive or other format. One example in the dataset is often called a case, instance or data point; in a table, a case is commonly represented by one row.

Each dataset represents only a selected and recorded portion of the real world. What was measured, who was included, how information was categorised, and what was omitted all affect what a model can learn.

Training data

Training data is the set of examples used to shape a model during training. It is the model's main experience of the problem.

A model cannot directly learn from parts of reality that never appear in its training data. Training data therefore influences both what the model can do and where it may fail.

Data quality

Data quality describes how suitable the available data is for the purpose for which it will be used. Relevant issues can include missing values, incorrect records, inconsistent measurement, outdated information, unreliable labels and poor coverage.

More data is not automatically better. A very large dataset can still be a poor learning resource if it is inaccurate, unrepresentative or badly matched to the intended task.

Data coverage / representativeness

Data coverage describes how well a dataset includes the kinds of people, situations, environments or cases in which a model is expected to operate. Representativeness asks whether important parts of that intended setting are adequately reflected.

A model may work well for common cases but poorly for groups or situations that appeared rarely, or not at all, during training.

Data provenance

Data provenance means knowing where data came from, how it was collected, how it was transformed, who created labels or categories, and under what conditions it may be used.

Provenance matters because the history of a dataset can explain hidden limitations. It can also raise questions about consent, ownership, cultural context and whether the data is appropriate for the intended use.

Data bias

Data bias is a broad term for systematic patterns in data that can lead a model to perform unevenly or reproduce undesirable historical relationships. Bias can arise from who is included, how things are measured, how labels are assigned, or how past decisions were made. One important form is selection bias, where the cases included in the dataset differ systematically from the wider population or setting the model is meant to represent.

Bias does not require malicious intent. A model can reproduce an undesirable pattern simply because that pattern was present in the data it was given.

Class imbalance

Class imbalance occurs when some outcome categories appear much more often than others in a classification dataset.

Imbalance matters because a model can appear successful by concentrating on the common class while performing poorly on the rarer class that may matter most.

Example

If 95 of 100 cases are "no disease", a system that always predicts "no disease" is 95% accurate yet fails to detect every actual disease case.

Generalisation and model fit

Generalisation

Generalisation is the ability of a trained model to apply what it learned to new cases that were not used to build it. Generalisation is the central reason machine learning is useful beyond the training dataset.

A model that performs well only on examples it has already seen has not demonstrated that its learned patterns will transfer to genuinely new situations.

The purpose of learning is not to become excellent at the training examples. It is to capture useful structure that remains useful on new cases.

Training set

A training set is the portion of the available data used to fit or shape the model. The model is allowed to learn from these examples.

A useful analogy is practice material: the learner is allowed to study it before being evaluated.

Test set

A test set is a separate set of examples held back from training and used to assess how the model performs on cases it did not learn from directly.

A useful analogy is an unseen exam. If the exam questions were already part of the practice material, the result would give an overly optimistic impression of learning.

Underfitting

Underfitting occurs when a model is too simple, insufficiently trained, or otherwise fails to capture enough of the useful structure in the data. It is one form of poor model fit: the learned representation does not capture enough of what matters for the task.

An underfitted model tends to perform poorly even on the examples it was trained on because it has not learned enough of the relevant pattern.

Overfitting

Overfitting occurs when a model becomes too specialised to the training examples, including quirks, noise or even memorised details that do not transfer to new cases. It is another form of poor model fit.

An overfitted model can achieve excellent training performance while performing noticeably worse on unseen data.

Example

A student who memorises the exact answers to practice questions but cannot solve a slightly different problem shows an analogous failure to generalise.

Evaluation and error

Evaluation

Evaluation is the process of judging whether a model performs well enough for its intended purpose. An evaluation metric is a numerical measure used to summarise one aspect of performance, but evaluation can consider more than a single score: it may examine which errors occur, whether performance differs across groups, whether the model remains reliable over time, and whether its speed, cost or interpretability are acceptable.

The most important question is not simply "Is the model good?" but "Good enough for what purpose, under what conditions, and with what consequences if it is wrong?"

Accuracy

Accuracy is the proportion of predictions that are correct overall. If a classifier makes 90 correct predictions out of 100 cases, its accuracy is 90%.

Accuracy is easy to understand but can be misleading when some classes are much more common than others or when different kinds of mistakes have very different consequences.

False positive

A false positive occurs when a model predicts that a condition or category is present when it is actually absent.

Whether a false positive is serious depends on the setting. The consequences of incorrectly flagging a legitimate bank transaction as fraud are different from the consequences of incorrectly flagging a medical test.

False negative

A false negative occurs when a model fails to identify a condition or category that is actually present.

In some applications, false negatives may be more costly than false positives - for example, missing a dangerous fault or failing to identify a serious disease.

Data leakage

Data leakage occurs when information that should have remained unavailable during training accidentally influences the model. The resulting evaluation can then look much stronger than the model's genuine ability to generalise.

Leakage breaks the logic of an independent test because the model has indirectly been exposed to information it was supposed to encounter only at evaluation time.

Example

It is like allowing a student to see information from the final exam while studying and then treating the exam score as evidence of completely independent learning.

When learned patterns can mislead

Distribution shift

Distribution shift occurs when the setting in which a model is used differs meaningfully from the setting represented in its training data. The people, behaviours, measurements, environment or underlying process may have changed.

A model can therefore have generalised well during testing and still become unreliable later or in another location.

Example

A fraud model trained before a new type of scam becomes common may lose effectiveness even though it performed well when it was first evaluated.

Correlation

Correlation means that two things vary together in a systematic way. When one changes, the other tends to change as well, although the relationship may be positive, negative or more complicated.

Correlation can be very useful for prediction. It does not, by itself, explain why the relationship exists.

Causation

Causation means that changing one factor actually contributes to changing another. Establishing causation requires stronger evidence than simply observing that two variables are associated.

Machine learning can make excellent predictions from correlations without knowing the causal mechanism behind them.

Spurious correlation

A spurious correlation is a relationship that appears predictive in the available data but does not reflect the meaningful relationship the model was intended to learn. It may arise from coincidence, a hidden third factor, background information, or quirks in how the data was collected.

A model does not automatically know which pattern humans consider the "right" one. If a shortcut is predictive, the learning process may use it.

Example

An image classifier intended to recognise an animal might partly rely on the typical background in the training images, such as snow, rather than on the animal itself.

Learning in the real world

Model lifecycle

A model lifecycle describes the sequence through which a machine-learning system is defined, built, evaluated, deployed, monitored and maintained. Deployment means placing a trained model into real use rather than examining it only in development or testing. A simple lifecycle is: define the task -> collect data -> train -> test -> deploy -> monitor -> re-evaluate or retrain.

The lifecycle view matters because a model that worked well when first released may become less reliable as data, behaviour or the surrounding environment changes. Monitoring checks whether performance remains appropriate; retraining updates or rebuilds the model using new or revised data when needed.

Human judgement moves upstream

Machine learning automates part of the process of constructing behaviour, but it does not remove human judgement. Human choices increasingly occur earlier in the pipeline.

People choose the problem, collect or approve the data, define labels, select the target, specify the objective, decide which errors matter, choose evaluation criteria, and determine where the model will be used.

The machine learns patterns, but humans shape what it has the opportunity and incentive to learn.

Week 2 - big ideas

These are the durable ideas to retain before moving into Week 3 on generative AI.

- Training shapes a model; inference uses the trained model on a new input.
- Data provides the learning experience, while an objective provides a direction for improvement.
- Different learning approaches obtain their learning signal in different ways: supplied labels, structure in the data, self-created prediction tasks, or feedback from actions.
- A dataset is a selected representation of reality, so coverage, quality, provenance, labels and imbalance can shape what a model learns.
- The central test of learning is generalisation: whether useful behaviour transfers to genuinely new cases.
- High training performance or high overall accuracy can be misleading when a model overfits, encounters imbalanced data, or benefits from data leakage.
- Predictive relationships are not automatically causal or meaningful; models can exploit spurious shortcuts that humans did not intend.
- The world changes, so a model that worked yesterday may require monitoring, re-evaluation and retraining tomorrow.
- Machine learning does not remove human judgement. It moves many important choices upstream into problem definition, data, labels, objectives and evaluation.